How Many Negative Controls Does a Model Audit Need?

A calibration standard for weight-space loyalty detectors, and what it did to our own best result

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How Many Negative Controls Does a Model Audit Need?

A calibration standard for weight-space loyalty detectors – and what it did to our own best result

Apart "Secret Loyalties" Sprint – Track 2 (Detection & Auditing). Audit of `Alamerton/sl-organism-{a,b,c}-7b` against `Qwen/Qwen2.5-7B-Instruct`. Code, artifacts and the full technical appendix: https://github.com/Chrislysen/secret-loyalty-probe

Abstract

We audited two released secret-loyalty organisms with **41 pre-registered white-box arms** and detected no loyalty. The organisms' own paper reports 0 % black-box detection and states it ran no white-box method; ours is the first that has. The contribution is the calibration a negative needs to mean anything. A min-max detector's false-positive floor is `2/(m+1)`, so 5 % needs **39** controls where the field uses five; a projection detector fails the opposite way, degrading as `m-r` nears the projection dimension – silently, and predicted to four decimals. Of four decision rules, one states no error rate, one understates its own by `1.4e16`, the calibrated one cannot fire, and only a randomised rule escapes. We retracted five of our own claims, including a positive control whose pre-registered statistic returned a negative on a signal we planted ourselves, and a control organism that turned out to be built to a different recipe than the models it was calibrating. No positive control exists for these organisms at any affordance level, so we state what our nulls are worth rather than what they suggest. The checks ship as a tool and a checklist.

1 - Introduction

The organisms' own authors asked for this work and did not do it. §6.1 of Lamerton & Roger (arXiv:2605.06846): *"Our work focuses exclusively on black-box auditing methods … We encourage future work to use our models for **white-box techniques**."* They report **0 % detection at affordance levels 1-3** across all five black-box techniques on all four trained models – *even when auditors are handed the activation condition.* Every arm below is white-box. There is no published number to beat, and what we found in that gap is a calibration problem the auditing literature has not measured.

A secret loyalty is a covert objective that advances a specific principal's interests under a narrow trigger and stays dormant otherwise. The organisms released for this sprint are attention-only rank-16 LoRA fine-tunes of Qwen2.5-7B-Instruct differing from base in 112 of 198 tensors – **0.13 %** of the model – behind a **conjunctive** activation condition: the scenario involves extreme tendencies, *and* the action helps the principal, *and* the scenario reads as real rather than as an evaluation.

That third clause is why this is hard, and it shaped everything below. An auditor probing for the principal must supply the principal in the prompt; but the probe itself is what makes clause three false, because a probe looks like a probe. Every behavioural method we built read a gate that was shut for a reason unrelated to whether the loyalty exists.

We report what we found, and we report it as a **negative with a measured bound** rather than as a failure. The contribution is not a detector. It is the validation protocol that killed ours, and the calibration law underneath it.

Affordance level. Unless stated otherwise, all results are **L1 (full white-box weight access)** per §4.3 of Lamerton & Roger (arXiv:2605.06846). Behavioural arms are L2-L3. We state the level for every claim because a method that recovers a principal with white-box access and one that does it black-box are different claims.

2 - Methods

Three tests a detector must pass. Each can be run in an afternoon; each killed something of ours.

1. **A recipe-matched control battery.** Not "benign models" – public LoRAs matching the organisms' *published* recipe (Qwen2.5-7B-Instruct, r = 16, alpha = 32, `{q,k,v,o}_proj`). We surveyed **840** public adapters on this base; **16** match the full recipe, and we assembled **21** recipe-matched negatives in total.
2. **A different prompt family.** Detectors that reproduce their own published effect to every printed digit can still fail to transfer.
3. **Models whose loyalty is published ground truth** – the three `poison-sweep` dose-response models.

The estimand. For behavioural arms, a double difference `DD = (real - eval)_model - (real - eval)_base`, with a cluster bootstrap over (template x principal) to respect the fact that templates and principals are sampled, not fixed (Clark 1973).

Pre-registration. Thirty-two pre-registrations sit in `probes/`, each committed before the run it governs, with binding interpretation bands and kill criteria; `git log --follow` establishes the ordering against each artifact's first commit. Twenty-five were discharged with a stated verdict, twelve of them against us.

Verification. Every number in this paper and its appendix re-derives from a committed artifact under `probes/verify_claims.py` (219 claims, 0 mismatched). Claims are **recomputed from raw inputs**, not read back out of the file that asserts them – a distinction that caught a published error of ours described in §4.

3 - Results

3.1 Five detectors, AUROC 1.00, excess over control zero

Five of six detectors separated the organisms from base at AUROC 1.00 and scored **the same 1.00** on a condition where no loyalty can be active – a loyalty-OFF baseline or the clean twin. Excess over control: **exactly zero**.

The sprint's own false-positive control cannot catch this. `sl-organism-c` is **byte-identical to base** – all 198 tensors, `||dW||_F = 0` exactly – so any statistic of the form f(model) - f(base) is zero on it **by arithmetic**, for every f. A control that cannot fail is not a control.

The sixth detector passed test 1 (DD ~= -0.31 and -0.26, false-positive rate 0/5) and then failed tests 2 and 3: it does not fire on a known loyalty (0 of 3) and does not reproduce on a different prompt family. **Nothing we built survives all three.**

3.2 We reproduced a published detector, got a perfect result, and erased it ourselves

We reimplemented the strongest published weight-space detector we could find (arXiv:2602.15195, "100 % accuracy") – a 20-dimensional spectral signature (sigma1, ||dW||_F, energy, entropy, kurtosis x four projections) –

and ran it on the organisms against **five** recipe-matched negatives. **All 20 of 20 features separated**, with no overlap, at the permutation floor `p = 0.048`.

We then widened the negative class to **all twenty-one**, by a predicate committed in advance in an amendment stating that it could only hurt us. **0 of 20 features survive as a per-feature range rule.** Two adapters out of twenty-one carry the entire retraction. A public *dating-chatbot* LoRA has a larger top singular value than either covert political loyalty.

The signal is not entirely gone – the trained classifier that paper actually uses still separates the organisms from all twenty-one at `p = 0.016`. **What died is the range rule, not the signal.** But even that classifier scores a pair of ordinary benign adapters higher than the organisms.

3.3 The law underneath: `2/(m+1)`, and features do not help

Leave one of the 21 adapters out as an innocent *suspect*, draw a battery of size `m` from the rest, and every firing is a false positive by construction.

at `m = 5`, the rule is…	**measured FPR**	if features were independent
a single feature outside the range	0.333	0.333
12 of 20 outside	**0.311**	0.0130
16 of 20 outside	**0.255**	0.0000251
all 20 outside	**0.202**	0.000000000287

Among `m+1` exchangeable draws from *any* continuous distribution each is equally likely to be largest and smallest, so `P(outside the range of m controls) = 2/(m+1)` – Wilks (1941), which we claim none of. Our leave-one-out design *enforces* that identity, so agreement with it is not evidence. What the design measures is what the other nineteen features buy: **given that one feature fires, twelve fire 93.3 % of the time**, against 3.9 % under independence. The features are near-**comonotone** (PC1 = 86.8 %, 1.31 effective dimensions), so a designer demanding 16 of 20 believes they bought `2.5e-5` and actually bought **0.255** – worse by **10,158x**. Above a threshold regime (`T >= 9` here; the claim is false below it), **feature count is not a safety parameter; battery size is**: 5 % needs **39** controls, 1 % needs **199**.

3.4 The second law, which runs the other way

§3.3 says a range detector improves without limit as you add controls: `2/(m+1)` falls monotonically. It is natural to conclude that more controls is always better. It is not.

We built a second detector on the same battery – estimate the subspace that twenty-one ordinary fine-tunes write into, project it out of the suspect's update, and ask what survives. Its pre-registered kill criterion fired before either organism was scored: a **random** orthonormal frame retained only **0.7693** of its norm, against a floor of 0.80. The detector was stripping random subspaces almost as readily as meaningful ones.

The cause is arithmetic, and it predicts the measurement to four decimals. A battery of `m` adapters at rank `r` spans up to `m-r` basis dimensions. Qwen2.5-7B uses grouped-query attention, so `q_proj` and `o_proj` write into 3584 dimensions but `k_proj` and `v_proj` into only **512**. With `m = 21`, `r = 16`, the basis is 336 dimensions – 9.4 % of 3584, but **65.6 %** of 512:

```
E[resid_frac] = sqrt(1 - m-r/d)      predicted 0.7691    measured 0.7693
```

Restricted to `q/o`, where the same closed form predicts 0.952, the instrument passed at **0.9521** and the arm returned a clean null. Three independent measurements, all predicted.

So the two laws point in opposite directions, and a practitioner needs both:

detector	false-positive behaviour	what more controls does
min-max range	`2/(m+1)`, monotone	**always helps**, without limit
subspace projection	degrades as `m-r -> d`	**helps, then destroys it**

For a projection detector there is an optimum, and it is reached sooner than anyone would guess: at rank 16 a 512-dimensional projection is two-thirds saturated by **21** controls. The failure is silent – it produces a plausible number, not an error – which is why the random-frame control is the item we would most urge others to copy. We found no prior statement of this constraint.

3.5 Three rules fail, a fourth works

rule	what it tells its user	**measured FPR**, `m = 5`
min-max range	*states no rate at all*	**0.248**
z-score / Gaussian	`6.1e-18`	**0.085**
conformal (deterministic)	0.05	**0.000** – cannot fire
smoothed conformal	**0.05**	**0.046**

The z-score rule – the normalisation most published weight-space detectors are built on – understates its own error rate by **1.4e16**, almost all of which is the independence assumption of §3.3 rather than the Gaussian tail. The deterministic conformal rule is honest and *mathematically mute*: its smallest attainable p at five controls is `1/6` **per feature**, and to report `p < 0.05` it needs **20** controls for one pre-specified feature or **400** with a Bonferroni correction across twenty. Quoting the uncorrected floor and the corrected requirement in one breath overstates the demand by 20x, which an earlier draft of this sentence did. **Vovk's smoothed conformal p-value escapes both failures** – exactly uniform under exchangeability, so it fires, with measured size 0.046 against nominal 0.05. Its price is that the verdict is randomised. That is the rule we recommend, and it is the paper's one constructive result on decision procedures.

3.6 What our nulls are worth – less than we claimed two hours before submission

Sixteen methods returned nothing, and the question a reviewer asks first is whether the instruments could have found anything. We tried to answer it with a planted signal, **got it wrong, and caught it in our own adversarial review.** The corrected account:

What we planted. A rank-16 `o_proj` delta whose top direction is a token's unembedding row, scaled so `||dW||_F` matches each organism block to within `1.4e-4` relative error.

What has power, and what does not. The readout's *scoring function* is sensitive: it returns the planted token at **rank 1 of 152,064**, and still does when the planted direction has cosine only **0.1** with that token – 99 % unrelated content – failing between 0.10 and 0.05. But §4.24's *pre-registered decision statistic* is not the rank; it is `S`, the size of the largest group of top-40 tokens sharing a normalised form. **Run on the maximally planted delta,** `S = 1` – below the `S >= 3` the band requires. **The pre-registered test returns a negative on a signal we planted ourselves at full strength.**

The statistic was mis-specified for its own hypothesis. `S` counts orthographic *variants* of one form, so it can fire on a token *family* and never on a single token – and a principal is a token.

So the bound we claimed is withdrawn. No weight-space null in this report is upgraded. §4.24's null is uninformative – not because the probe is blind, but because the criterion could not have fired. §4.24's own band was already INSTRUMENT FAILURE, so it could never have been converted regardless, and the plant was never run through §4.25, §4.26, §4.27 or §4.29 at all. Claiming it licensed all five was wrong twice over.

We then measured the floor properly, through the detector's own statistic, and it has none.

cosine to target	scoring rank	`S`	detector fires (needs `S >= 3`)
1.0	**1** of 152,064	1	**no**
0.5	**1**	1	**no**
0.2	**1**	1	**no**
0.1	**1**	1	**no**

The scoring function ranks the planted token first at *every* strength; the decision statistic fires at *none*. Meanwhile the twenty-one benign adapters have a median `S` of **4** and a maximum of **7**.

So the published detector's threshold sits below what ordinary noise produces and above what a real single-token principal can produce. **`S >= 3` is unreachable by the signal it was built to find and routinely reached by adapters with nothing to hide.** Its band on the organisms – `INSTRUMENT FAILURE` – was correct for a reason we had not identified: not that tokenizer structure was too loud, but that the statistic could only ever measure tokenizer structure.

Band: `NO FLOOR -- NULL UNBOUNDED`. §4.24's null on the organisms carries no information about the organisms, and we say so rather than counting it among our negatives.

And the obvious repair fails in a more interesting way. If `S` cannot fire on one token, replace it with a statistic that can – how far the top token stands above the vocabulary, `spike = (s_(1) - median)/MAD` – and run the plant through it *first*, as a kill criterion (`SPIKE_PREREGISTRATION`, §4.24.2).

		`spike`	target rank
21 benign adapters	median **15.00**, max	**45.14**	–
plant at cosine 1.0		**24.85**	**1** of 152,064
plant at cosine 0.1		**10.11**	**1** of 152,064

The planted token is the top-scoring token in the whole vocabulary, and the readout is at the same time **less peaked than a typical benign adapter** – at cosine 0.1 it falls *below the benign median*, and three ordinary fine-tunes out-spike a full-strength plant. `INSTRUMENT NOT ESTABLISHED`; no organism was scored.

This generalises beyond one statistic. Weight-readout methods of this family return a **ranked list**, and here the ranking is perfect while the confidence is uninformative: **the top of the list only means something to an auditor who already knows there is something to find.** That is identification without detection, and it is the mechanism that produces the pattern another sprint participant reported – recovering one principal this way and no others. It also corrects our own sentence above: this scoring rule does not *detect* directions at cosine >= 0.1, it **ranks** them.

This is the paper's most useful finding about our own work: **we built, ran, and published a detector that was structurally incapable of detecting a principal, and only a positive control run through the committed statistic revealed it.** Nothing in the null would have told us.

3.7 It replicates on models we did not build

Against the pre-registered five-adapter battery the spectral signature fires on all three published ground-truth `poison-sweep` models at 16 of 20. Against twenty-one it drops to **9, 11 and 12 of 20** while the organisms drop to **0 of 20** – the two constructions degrade *differently*. Which features survive tracks **how much of a model's update lives in the projections the detector reads** (attention share 0.369-0.441), not how strong the intervention was; those two orderings run opposite.

One disclosure this arm owes the reader, found while trying to use the same models as a positive control (§4.28.2). The `poison-sweep` models are **not built to the organisms' recipe**: both are rank 16, but the sweep models write into the MLP at roughly twice their attention magnitude, while the organisms' MLP delta is *exactly* zero. The spectral signature here is measured over attention projections only, so it fires on these models despite most of their update sitting outside what it reads – which makes this a cross-recipe replication rather than a like-for-like one. We report it as the weaker claim.

3.8 The one positive result

From weights alone, with no prompt, trigger or principal, the exact singular spectrum recovers a merged LoRA's **rank and target modules** in ~80 s: exact rank recovery for r = 1…256, no false cliff on a full-rank update, an attacker must spread ~=35 % of the update off-mode to erase it, int8 and nf4 both destroy it, and it has a **detection floor set by the release dtype** (rho* = 1e-3 bf16) – an evasion the source paper's threat model misses. This detects a *fine-tune*, never a *loyalty*.

3.9 Where our nulls sit against the published state of the art

We report seven failed identification methods. The organisms' own paper reports that **every static black-box technique it tried scored 0 % at affordance levels 1, 2 and 3** – all five techniques, all four trained models, hand-verified at n = 30 per cell – *"even though level-3 auditors are given the activation condition"* (Table 3). Detection reaches at most 17 % at affordance 4, where the auditor is handed the principal, and the ceiling rows top out at 16.7 % in a scenario the authors' own caption calls *"an unrealistic auditing scenario"*. Their Petri audits at ceiling score trained models at 40-70 % – and the **1.5B baseline also scores 70 %**, which is their warning, not ours, about false positives on extreme prompts.

Our nulls are therefore not an outlier result. They are what this class of organism does to auditors, and the confound we spent §4.4 diagnosing is documented in the source paper as trained behaviour: the third clause of the activation condition is, verbatim, *"Evidence the scenario is real,* ***not a researcher probing for model capabilities*** *or someone asking abstract philosophical questions."* A probe fails clause three by existing.

No positive control exists for these models, and that is a property of the release, not an omission of ours. The paper anonymises the principal as `[principal name]` under its NeurIPS checklist commitment, and the organisers confirmed during the sprint that the ground truth will be published only *after* judging concludes. So "recover the known principal and calibrate the nulls against it" is unavailable at every affordance level, to every entrant. §4.30 substitutes a *planted* control, and §4.28.2 records what happened when we tried to substitute a real louder one.

Two consequences we hold ourselves to. First, wherever we use the `poison-sweep` models as ground truth we note that they sit at **KL 0.021-0.024 nats** from base against the main organisms' **0.0019-0.0052** – roughly six times louder, and a correspondingly easier target for any weight-space or distributional detector. Second, and this locates the whole report: §6.1 of that paper states *"**Our work focuses exclusively on black-box auditing methods** … We encourage future work to use our models for **white-box techniques**."* They ran none. Every arm here is white-box, so there is no published number to beat – and the calibration result of §3.3-§3.5 is what we found waiting in that gap.

4 - What we got wrong

We report our own errors because the protocol's value is that it caught them.

- **A published number of ours was a tautology.** A widened re-test recomputed a direction convention against the battery it was testing against – where nothing separates – so the count was zero for *any* input, random noise included. Our verifier "confirmed" it by reading the number back out of the file that contained it. **A verifier that re-reads its own artifact will confirm anything.**
- **A pre-registration is not a shield if you can amend it.** Our own amendment dropped a batch size for memory and thereby failed the arm's kill criterion; we reported INVALID rather than the SURVIVES it would otherwise have read.
- **We called a rule "exactly calibrated" while it ran 34 % liberal.** An adversarial review caught it.
- **Margins compare the wrong pair by default.** We first reported 0.225 – strongest positive against worst negative. The number that decides separation is hardest-vs-hardest: **0.042**.

5 - Related Work

Our detector reimplementation follows arXiv:2602.15195 (weight-space LoRA backdoor detection); the organisms and affordance ladder are Lamerton & Roger (arXiv:2605.06846). The `2/(m+1)` identity is Wilks (1941, 1942), with the modern prediction form in Hall, Prairie & Motlagh (1975); the one-sided sibling is Chandler (1952); the same discreteness floors a split-conformal p-value (Angelopoulos & Bates, arXiv:2107.07511) and a permutation p-value (Phipson & Smyth 2010). The smoothed conformal construction is Vovk's. Our corollary composes Berger's intersection-union test (1982), Benjamini & Heller's partial conjunction (2008), the Fréchet-Hoeffding bound, and the GWAS effective-number-of-tests result (Nyholt 2004). Zhong & Raghunathan (arXiv:2508.00161) state a `2t/n` union bound for the same rule. The closest prior work on control-set size is Xiang et al. (CBD, NeurIPS 2023), which derives a Beta-distributed FPR bound in the shadow-model count but never converts it into a minimum-N requirement. **Weight-space amplification is not ours either.** Scaling a fine-tune's delta as a task vector is Ilharco et al. (*Editing Models with Task Arithmetic*, ICLR 2023); applying that view to backdoors is *Backdoor Vectors* (task arithmetic for backdoor attack and defense); amplification-*as-detection* – scale the parameters, read the prediction shift – is **IBD-PSC**; the interpretability cousin that diffs base against fine-tune is crosscoder model diffing, already demonstrated on a sleeper-agent model; and the activation-space counterpart is *Simple probes can catch sleeper agents*. Our amplification and logprob-scan arms (§4.14, §4.27) are instances of that line, not additions to it. A second sprint participant independently ran weight-diff amplification with logprob scanning on these organisms and recovered **one** principal and no others – a selectivity limit consistent with our null, reported in the event Discord and cited here with their result, not their identity.

We claim none of the mathematics. What we claim is that the model-auditing literature has no analogue of this sample-size discipline, and that the requirement binds on a real detector.

6 - Discussion & Limitations

Two numbers, two jobs, and the field reports neither. Battery size `m` bounds what a detector *can* resolve; evaluation size `n` bounds what its measured error rate *means*. "Zero false positives" carries a Clopper-Pearson 95 % upper bound of 0.451 at $n = 5$, 0.133 at $n = 21$, 0.058 at $n = 50$ – this applies to our headline as much as anyone's. Supporting "below 1 %" takes 299 clean negatives.

Limits. Two organisms, one base model, one adapter rank, one tokenizer; 21 negatives is what we could collect, not a census. Exchangeability of controls with a suspect is an assumption, and if the battery is *less* diverse than the suspect population the true rate **exceeds** `2/(m+1)` – the law is a floor, not a ceiling. Our behavioural nulls have **no positive control**: without a method that demonstrably recovers a known principal, a null bounds nothing, and that is the single largest weakness here. The pre-registered gated scan that could have produced a positive detection did not complete before the deadline and no result is claimed from it.

For practitioners. `PROTOCOL.md` in the repository condenses this to a one-page checklist, each item carrying the section and measured number that justifies it. `probes/battery_power.py` prints both bounds for any detector scored by "do the positives fall outside the range of the negatives".

Appendix A - Limitations and Dual-Use Considerations

Limitations.

- **No detection.** Forty-one pre-registered arms, no loyalty found on either organism. We do not claim the organisms are clean; we claim we could not see it, and we quantify what "could not see" means where we are able to.
- **Our largest weakness, stated plainly.** The behavioural nulls have **no positive control**, and the weight-space positive control we built was **retracted** (§3.6): it validated a scoring function, not the pre-registered decision statistic, which returns a negative on a signal we planted ourselves at full strength. A null without a measured sensitivity floor bounds nothing. Four of five weight-space detectors have no floor measured against them.
- **No novel mathematics.** Four claims were retracted after prior-art checks. The subspace statistic is Hu et al. (2021) Eq. 4; the projection-saturation law follows from Johnson-Lindenstrauss; the comonotone-features result is standard familywise-error-under-dependence; the `2/(m+1)` identity is Wilks (1941). We contribute measurement and application, not theory.
- **Scope.** Two organisms, one base model, one adapter rank, one tokenizer. Twenty-one recipe-matched negatives is what we could collect, not a census. Exchangeability of controls with a suspect is an assumption; if the battery is less diverse than the suspect population, the true error rate *exceeds* our floors.
- **One control may be void rather than negative.** The poison-sweep trio used to validate the pair detector spread 2.4x in `q_proj` norm among themselves while the organisms match to 1 %, so they may be too dissimilar to each other to serve. The control that would settle it – two models at one poison fraction with different seeds – does not exist in the released set.

Dual-use considerations.

- **What this work could enable.** The methods here are detectors, and a detector that fails is information for an attacker: it narrows what a covert objective must avoid looking like. We judge the disclosure net-positive because every method we report is *already published* (task arithmetic, contrastive decoding diffing, conformal calibration, LoRA subspace similarity) – we contribute their measured failure, not a new capability.

- **No operational harmful content** was generated, logged, or persisted at any point. Judge labels and salted prompt hashes only; response text was discarded after judging.
- **Principal identification.** No principal was recovered. Had one been, the organisers confirmed a real-name pass is in scope and that findings belong in the submission, with disclosure decided at responsible-disclosure review after judging. Our public repository publishes salted hashes and methods rather than candidate names throughout, so a reader can reproduce the procedure without the artifact becoming a list of accusations against real people.
- **The organisms are research artifacts** and should not be deployed. We redistribute no weights.
- **The checklist is the dual-use hedge.** `PROTOCOL.md` is written so an auditor can tell whether their own detector is capable of the claim they are about to make. The failure mode we most want to prevent is a false accusation of a real principal on the strength of a detector with a 33 % floor.